

Project Progress Report – ECG System Development

Artificial Intelligence Development

The first phase of the project involved the use of AI techniques for ECG analysis. A convolutional neural network (CNN) was developed and trained using ECG datasets to classify cardiac conditions such as normal rhythm, tachycardia, bradycardia, and possible atrial fibrillation. Data preprocessing, feature extraction (including R-peak detection), and model evaluation were performed to ensure reliable classification performance.

Embedded System Design

Following the AI development, the system architecture was designed using an ESP32-S3 microcontroller as the main processing unit. The hardware includes:

- AD8232 for ECG signal acquisition
- ADS1115 for analog-to-digital conversion
- OLED display for real-time visualization
- USB-C interface for power supply

Firmware development included signal acquisition, processing, and display control, ensuring integration between hardware and software components.

PCB Design

Due to the delay in receiving the physical components, a custom PCB was designed to optimize the system layout and serve as a future improvement of the prototype. The PCB design includes:

- Proper component placement to minimize noise in the ECG signal
- Separation of analog (AD8232) and digital (ESP32) regions
- Power routing and I2C communication lines
- Consideration of grounding strategies and signal integrity

This step was also used as an opportunity to improve hardware design skills and prepare for a more compact and robust final version of the device.

Current Status

At the moment, the project is awaiting the arrival of all hardware components. In the meantime, the PCB design was developed both as a productive use of time and as a planned feature for future iterations of the system. I would like to ask you Prof. Latorre to check the zip file with the KiCad PCB and schematics files to check if everything is correct or if it needs alteration. Thank you.